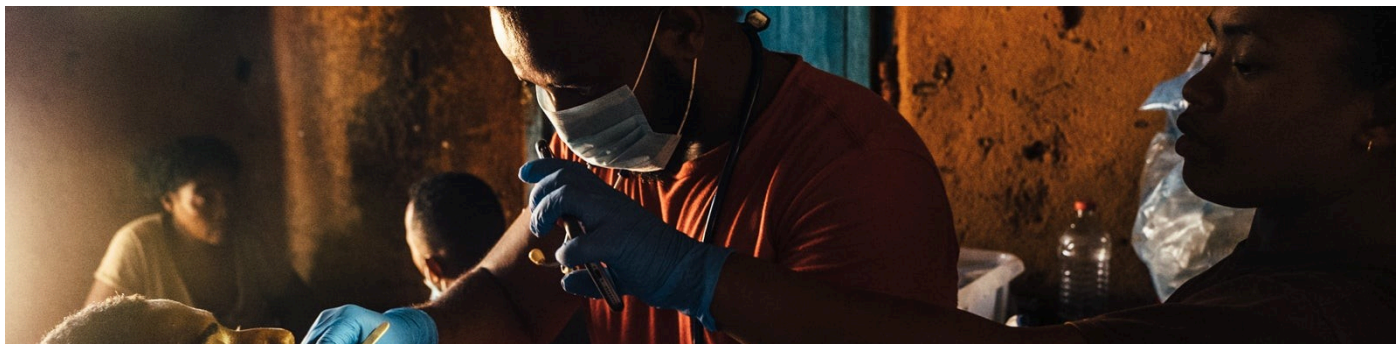




## Climate-Smart Public Health: A Blueprint for Resilient Health Systems

### Abstract

In a 2025 article in *The Lancet Planetary Health*, Harvard Chan researcher Christopher Golden introduced Climate-Smart Public Health, a data-driven framework built on four pillars: surveillance, risk assessment, early warning, and resilient health infrastructure. Using Madagascar as a proving ground, the article offers a blueprint that governments anywhere can adopt to protect people from climate-related health threats.

### Introduction

Extreme weather, food insecurity, and the spread of disease are becoming more common, so countries need public health systems that can adapt. Madagascar faces high rates of poverty, malnutrition, and maternal mortality, along with frequent cyclones, floods, heatwaves, and droughts. That combination made it a critical place to launch a new approach. Golden's framework, Climate-Smart Public Health (CSPH), connects health data with climate and environmental data and turns the results into action. It combines input from local communities, health workers, and organizations, and it is designed to be adaptive, bottom-up, and flexible as climate hazards change. The goal is a concrete, scalable blueprint for climate-resilient health systems.

## How Do Scientists Study This?

Golden's team put CSPH into action through two projects in Madagascar: monitoring droughts to predict crop failures and malnutrition, and detecting harmful algal blooms to watch for marine food poisoning. They merged health data from thousands of clinics with climate and environmental data, including satellite measurements of temperature, rainfall, and crop status. They used artificial intelligence and satellite imaging to monitor and predict droughts and algal blooms. Working with Madagascar's Ministry of Public Health, other ministries, and NGOs, they built databases that can share information. Finally, they created high-resolution, time-specific risk maps and early warning alerts to guide real interventions.

### The four pillars of Climate-Smart Public Health

#### 1 Surveillance

Link health, climate, and sensor data

#### 2 Risk assessment

Map who is most at risk

#### 3 Early warning

Alerts through digital and radio

#### 4 Resilient clinics

Solar power, trained providers

#### Merge the data

Records from thousands of clinics were combined with satellite climate and crop data.

#### Predict with AI

Artificial intelligence and satellite images forecast droughts and harmful algal blooms.

#### Map and alert

Risk maps and early warnings guided food aid and health resources to communities.

## What Did They Discover?

The article organizes CSPH into four pillars. Surveillance and monitoring link routine health data with open climate data and low-cost sensors to fill gaps. Risk assessment models how hazards like drought lead to lower harvests and malnutrition, and maps who is most at risk. Early warning and response use forecasts, satellites, and health data to alert officials and communities; digital and radio alerts work well in low-resource settings. Resilient infrastructure and provider capacity means flood-resistant, solar-powered clinics and training health workers to recognize climate-sensitive diseases. In Madagascar, early alerts helped target food aid and health resources to affected communities, showing how the approach can work where resources are limited.

### CSPH in Madagascar

**4**

pillars in the Climate-Smart Public Health framework

**1000s**

of health clinics' data merged with climate data

**2**

pilot projects: drought and harmful algal blooms

Even with limited data, combining local knowledge with available satellite technology can reveal critical risk patterns.

## Takeaways

### Four pillars

Surveillance, risk assessment, early warning, and resilient infrastructure work together.

### Data drives action

Merged health and climate data led to alerts that targeted real aid.

### A scalable blueprint

Strategies tested in Madagascar can be adapted for health systems worldwide.

## Key Terms

### Framework

An organized structure of ideas and steps that guides how to approach a complex problem.

### Remote sensing

Collecting information about Earth from a distance, usually by satellites or aircraft.

### Risk assessment

Estimating how likely a hazard is to cause harm and who is most likely to be affected.

### Interoperable database

A data system designed so different organizations can share and combine information easily.

### Climate-sensitive disease

An illness whose spread or severity changes with weather and climate, such as malnutrition or diseases spread by mosquitoes.

## Words to Know · Climate-Smart Public Health

Framework

Remote sensing

Risk assessment

Interoperable database

Climate-sensitive disease

## Activity

### Four-Pillar Plan

Students apply the four pillars of Climate-Smart Public Health to a climate hazard in their own region.

Materials: Four-pillar planning template; Local climate hazard information; Markers or slide tool

1. Choose a climate hazard that affects your region, such as heat waves, floods, or wildfire smoke.
2. Surveillance: list the health and climate data you would need to track.
3. Risk assessment: identify who is most at risk and where.
4. Early warning: design an alert, including who receives it and how, such as text or radio.
5. Infrastructure: propose one upgrade for a local clinic or school, then present your plan.

## References

Miller, A. (2025). Climate-Smart Public Health: A blueprint for resilient health systems. Harvard Chan C-CHANGE.

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Golden, C. D., et al. (2025). Climate-Smart Public Health. The Lancet Planetary Health, as described in the article above.